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CO₂-Based Plastics for Downhole Water Shutoff and Alternative Sequestration Integrated Process

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Abstract

CO₂-based plastic material uses CO₂ as feedstock instead of crude oil for plastic manufacturing to reduce the CO₂ footprint. This work introduces CO₂-based plastics into upstream operations for both downhole water shutoff and CO₂ alternative sequestration purposes in fractured reservoirs. The performance was identified and optimized by investigating the blocking and packing behaviors of CO₂-based plastic particles in different fractures.

A total of six CO₂-based plastic samples, in which CO₂ contents ranged from 15% to 40%, were evaluated and compared. All the samples were grinded into similar particle sizes 40-80 mesh. The blocking ability was tested by injecting brine into a CO₂-based plastic-packed open fracture. The packing behavior was quantified by injecting a CO₂-based plastic suspension into a contraction fracture. The pressure buildup during flow was monitored in both processes.

The CO₂-based plastics performed quite differently from the commonly used water-swelling polymer gels in fractures due to the non or slightly deformable nature of the plastics. In the blocking test, stable differential pressures linear with flow rates were observed when water flowed through the plastic pack in the open fracture, while the pressure responses to flow rates were usually non-linear for the polymer gel pack. Two CO₂-based plastic samples, T2 and T9, produced much higher pressure to water flow than the other four samples at 95°C. Samples with low glass transition temperature partially melted at 95°C and showed very weak blocking. In the packing tests, CO₂-based plastic suspensions demonstrated weaker deformation and passing abilities in the contraction fracture than polymer gel, resulting in higher packing pressures. T2 and T9 suspensions showed much quicker packing and pressure buildup than other samples. The blocking capability of the in-situ formed pack to the following water flow was similar for the two samples at a fixed packing pressure. At a high temperature of 95°C, the packing was delayed with a weak blockage to the subsequent water flow. Increasing particle size was an effective way to solve this problem.

This work developed a water shutoff method integrated with CO₂ alternative sequestration by using CO₂-based plastics. Some insights were provided on the performance of materials with different deformation abilities in fractures.

Introduction

CO₂ is a significant greenhouse gas that traps heat in the Earth's atmosphere, contributing to global warming and climate change. As an inexpensive, and readily available carbon source, extensive efforts have been made globally to capture and convert CO₂ into useful chemicals for carbon neutralization. One of the most promising applications for captured CO₂ is the production of CO₂-based plastics, which use CO₂ as a feedstock for plastic manufacturing, replacing crude oil and natural gas. In this process, CO₂ is utilized and fixed into solid polymers rather than being directly sequestered underground, thereby helping to reduce the CO₂ footprint. Because CO₂ is quite thermodynamically stable, catalytic reaction with highly reactive reagents such as epoxides is needed in the synthesis. In 1969, Inoue et al. successfully copolymerized CO₂ with propylene oxide (PO) using a diethyl zinc/water catalyst system, creating the first polypropylene carbonate (PPC). Over the past 50 years, PPC has undergone extensive research and development and became the most successful commercialized CO₂-based plastic products.

PPC is recognized as a biodegradable polymer due to the presence of ester groups (-COO-) in its backbone, which are prone to hydrolysis (Ye et al., 2019; Li et al., 2022). The polymer's terminal hydroxyl groups (-OH) can lead to unzipping-type degradation at elevated temperatures, resulting in thermal instability with a decomposition temperature below 220°C (Li et al., 2022). Conventional PPC typically exhibits a low glass transition temperature (T_g) below 40°C and lacks mechanical strength. This is primarily due to the short methyl side groups (-CH₃), the flexible ether bonds (-O-) in the main chains, and its amorphous structure (Qin, et al., 2010; Liu et al., 2010; Li et al., 2022). These characteristics have limited its practical applications. Table 1 compares the properties of PPC with those of another biodegradable material, polylactic acid (PLA).

Table 1—Basic properties of PPC and PLA

Property	T _g , °C	Tensile strength, Mpa	Density, g/cm ³	Burning heat, 10 ³ KJ/Kg	Td, °C
PPC	30-41	33.2	1.275	18.5	218
PLA	55-60	40-55	1.23-1.27	19.0	315

PPC has been strengthened by various technologies, including increasing molecular weight by novel catalysts (Zhu et al., 2002; Xu et al., 2014; Ye et al., 2019), ternary copolymerization by introduce rigid or crystalline unit into the backbone (Wang et al., 2022), block copolymerization using hydroxyl-terminated PPC and other reactive blocks (Luo et al., 2017; Zhang, et al., 2020). Additionally, physical blending with other polymers or inorganic fillers, and crosslinking PPC with crosslinkers, have been employed (Li et al., 2022; Bora et al., 2023). Among these modifications, PPC-based polyurethane stands out as a practical and commercially successful example (Harris et al., 1990; Wang et al., 1999; Ma et al., 2018). PPC polyol, a low molecular weight PPC terminated with reactive groups, is synthesized from the copolymerization of CO₂ and PO. It serves as the soft segment to react with diisocyanates, producing PPC based polyurethane. PPC-based polyurethane exhibits glass transition temperatures ranging from 40-60°C, which are significantly higher than those of PPC. Additionally, it has been reported to possess superior hydrolytic stability, antistatic properties, and antioxidant capabilities compared to polyester-based polyurethane. It combines the eco-friendly of PPC and elastic of polyurethane, making it suitable for various applications.

CO₂-based plastics have potential applications in water control technology. The most common material currently used for water control is preformed particle gel (PPG) (Han et al., 2014; Bai et al., 2015; Seright and Brattekas, 2021; Cao et al.). PPGs are superabsorbent polymeric materials that become soft and deformable in aqueous solutions after absorbing water. This property allows them to migrate easily and pack tightly in fractures, effectively blocking water flow. However, due to their low strength, PPGs are prone to rupture and dehydration under pressure gradients from fluid flow. In contrast, CO₂-based plastics are

rigid materials that can potentially withstand flushing without mechanical deformation. Some studies have explored the use of plastic particles for conformance control (Li et al., 2019). However, the thermoplastic nature of these materials and their tendency to decompose at elevated temperatures present significant challenges. The objective of this study was to assess the feasibility and enhance the performance of CO₂-based plastics for downhole water shutoff applications. This involves integrating CO₂-based chemicals into upstream operations to create a combined water shutoff and sequestration process. Various commercially available CO₂-based plastics were collected and grinded into particles. These particles were then evaluated for their blocking and packing abilities in different fracture geometries.

Experimental

CO₂ based plastic material

Total 6 CO₂ based plastic samples were collected from local market. The products are all transparent resin with initial particle size 5-10 mm. Table 2 lists the basic properties of the 6 samples. For samples T2, T4, and T9, they are PPC based polyurethane, so the CO₂ contents of the three samples were lower than other three PPG samples.

Table 2—Basic properties of CO₂ based plastic samples

Performance Index	T4	T2	T9	C1	C2	TG
Appearance	Transparent Resin					
Density, g/cm ³	1.25±0.05			1.27±0.05		1.20-1.35
Glass Transition Temperature (T _g), °C	45	56	56	23	29	20-30
Melting Temperature (T _m), °C	195	195	195	-	-	-
Tensile Strength, Mpa	13	13	32	33	33	5
CO ₂ content, Wt. %	~15	~30	~25	40	40	40
Structure	PPC based polyurethane			PPC		

Brine

The brine was synthetic injection water with salinity 2,425 mg/L.

Table 3—Ion composition of injection water

Brine \ Ion, mg/L	Na ⁺	Ca ²⁺	Mg ²⁺	Cl ⁻	HCO ₃ ⁻	SO ₄ ²⁻	TDS, mg/L
Injection water	442	262	80	760	199	682	2,425

Physical grinding

The CO₂ based plastic samples were physically grinded into different mesh sizes using liquid nitrogen mill technology. Samples with mesh size between 40 mesh and 80 mesh were used in the tests.

Thermal stability test

The long-term thermal stability was tested by aging CO₂ plastic in brine at 95°C. Put 1 g CO₂ plastic particles and 30 mL brine into Penicillin bottles. The brine was removed O₂ by N₂ bubble method for 2 hours in the glove box. Seal the bottles in the glove box and transfer into oven at 95°C. Monitor the weight for maximum 6 month.

Procedure of blocking test

The CO₂ based plastic particle blocking tests were conducted by the core flow method. An artificial fracture metal plug made of stainless steel were used to eliminate the effect of the matrix permeability. The fracture size is 7.0cm*2.0cm*0.2cm (L*W*H of 2.76 inch*0.79 inch* 0.08 inch).

The procedures are as listed below:

1. Put quantitative dry CO₂ base plastic particles into the fracture to form a CO₂ based plastic pack.
2. Load the fracture model packed with plastic into coreholder and set confining pressure to 1000 psi.
3. Increase the temperature to 95°C.
4. Inject brine into the core at flow rate of 0.5, 1.0, 2.0, 4.0, and 8 ml/min and record the differential pressure across the pack.

Procedure of packing test

The CO₂ based plastic packing tests were conducted by the coreflooding method. An artificial contraction fracture metal plug was used. The fracture size is 7.0cm long and 2.0cm high. The width changes from 0.2cm to 0.025cm. Diagram sketch of the fracture setup is shown in Figure 1. The CO₂ based plastic particle suspension in an accumulator was pumped out to the fractured metal plug. The pressure buildup along the fracture was monitored. A magnetic stir bar inside of the accumulator was used to disperse the particles during the pumping. The inner diameter of connection line was 0.5 cm (~0.2 inch) to ensure free flow of particles through the lines.

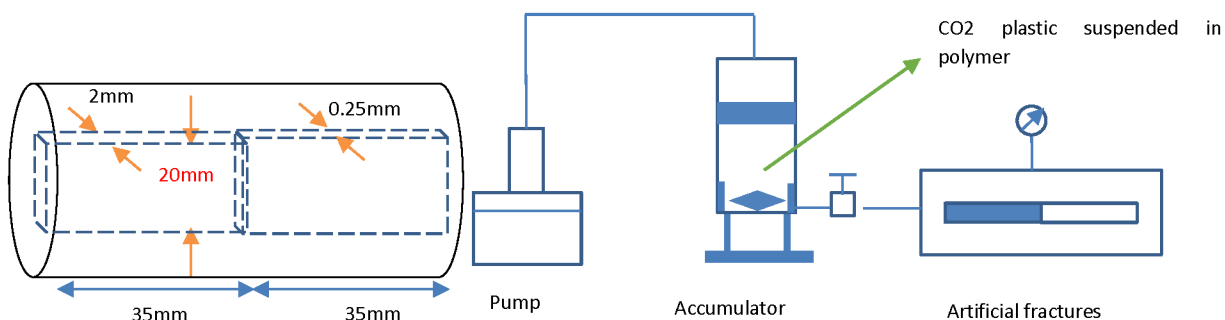


Figure 1—Diagram sketch of the fracture setup

The procedures are as listed below:

1. Prepare CO₂ based plastic particles suspension in brine by stirring.
2. Add polymer powder to the suspension and stir the mixture at 800 r/min for 20 mins.
3. Put the mixture into the accumulator with magnetic stirring.
4. Connect the accumulator and the fracture setup using lines with inner diameter of 5mm.
5. Inject CO₂ based particle suspension into the fracture at flow rate of 10 ml/min. During the injection, stir the sample in the accumulator at 200 r/min. Record the pressure along the fracture. Stop the injection when pressure arrived at 4 Mpa (~580 psi)
6. Switch the injection to brine at flow rate of from 1 mL/min to 8 mL/min and record the differential pressure

Results and discussion

Thermal stability

CO₂ based plastic is a biodegradable material. The applications of this material were unusually for the daily packing or agricultural film exposed in air as an environment friendly alternative of normal plastic.

Differently, the reservoir is a high temperature liquid environment without air. The thermal stability test evaluated the long-term stability of CO₂ based plastic in brines at elevated temperature. Figure 2 shows the weight reservation of different samples after aging at 95°C. For the polyurethane samples, T2 and T4 did not show any weight loss after aging in brine at 95°C for 3 months. T9 started to show weight loss after 1 month aging in brine. For PPC samples, C1 showed weight loss after aging in brine at 95°C while C2 was relatively stable. After more than 6 months, T2 and T4 did not show significant weight loss with no appearance change, T9 reserved 80% weight and the shape did not change, C1 reserved 70% weight and became fragile, C2 reserved 90% weight and melted together. Most of the samples showed good thermal stability in brines at 95°C comparing with the cases used on ground. The possible explanation is the absence of O₂ in the brines.

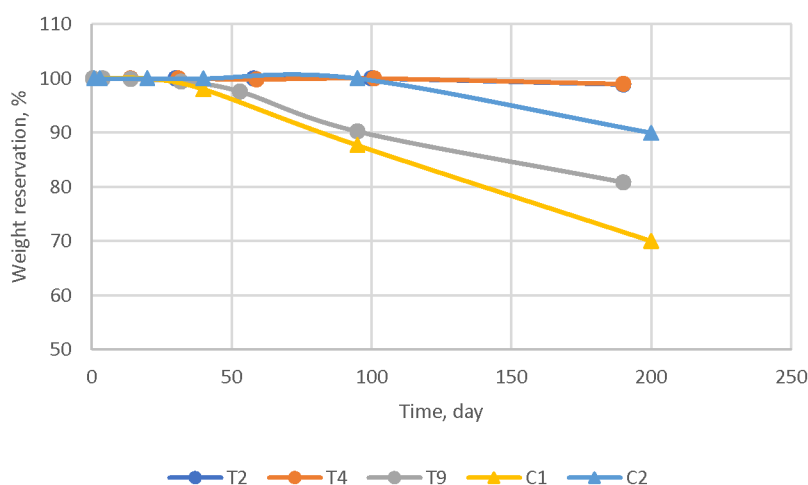


Figure 2—Thermal stability of CO₂ based plastic samples

Physical grind

Initial samples of CO₂ plastic were too large to be injected into reservoir. Physical grind is necessary before injection. Normal physical crush method produces heat that could melt the plastics to liquid like, so liquid nitrogen crush technology was used to cool down the samples and keep the brittleness. Figure 3 compares the sample before and after grinding. Grinded samples with mesh size between 40 mesh and 80 mesh were used in the tests. Table 4 lists the basic properties of the crushed samples. All the samples were oil-wet with contact angle to oil ranged from 56° to 75°. The apparent density of crushed T2 and T9 was much lower than other 4 samples.

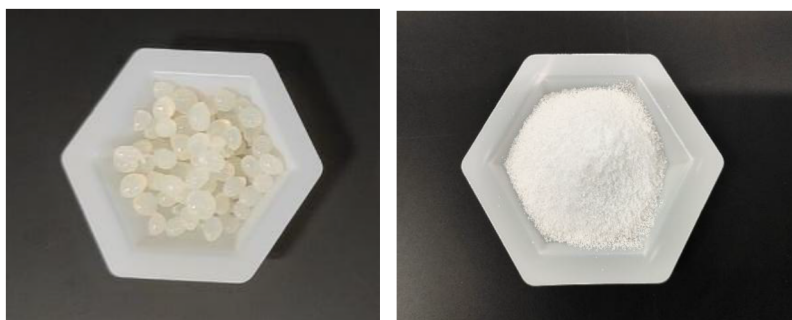


Figure 3—CO₂ based plastic T2 before and after grinding

Table 4—Properties of grinded samples

Sample	Size, mesh	Absolute density, g/cm ³	Apparent density, g/cm ³	Contact angle to oil, °	Solid content, %
T2	40-80	1.2004	0.3921	74.46	100
T4	40-80	1.2038	0.6757	67.47	100
T9	40-80	1.1952	0.4166	63.54	100
C1	40-80	1.2934	0.6409	57.95	100
C2	40-80	1.2476	0.6298	56.29	100
TG	40-80	1.2292	0.6670	73.46	100

Blocking test

The blocking tests evaluated the blocking capabilities of CO₂ plastic particles after packing in the open fracture. The particle size was 40-80 mesh and the packing weight was 2.65g. Figure 4 shows pressure buildup during water flow through CO₂ plastic packed fracture at 95°C using sample T4 as example. Stable pressure was produced by the plastic pack in the fracture even at flow rate as high as 8 mL/min. The stable differential pressures were proportional to flow rates. In comparison, the pressure produced by preformed particle gel (PPG) became not stable at flow rate 4 mL/min and the pressures and flow rates were non-linear. However, the pressure values of tests using CO₂ based plastic were much lower than that of PPG. Plastic material was not deformable or slightly deformable at temperature much lower than the melting temperature. And usually, the deformation was permeant with no elastic. So, there will be interparticle pores in the pack formed by CO₂ plastic with relatively high permeability. Correspondingly, the pressure during water flow was low. PPG is a water absorbing material and easy to deform. On one hand, the deformation can partially fill interparticle pores and make the pack low permeability and high-pressure buildup. On the other hand, the pack will become unstable at high flow rate due to the movement by water flush.

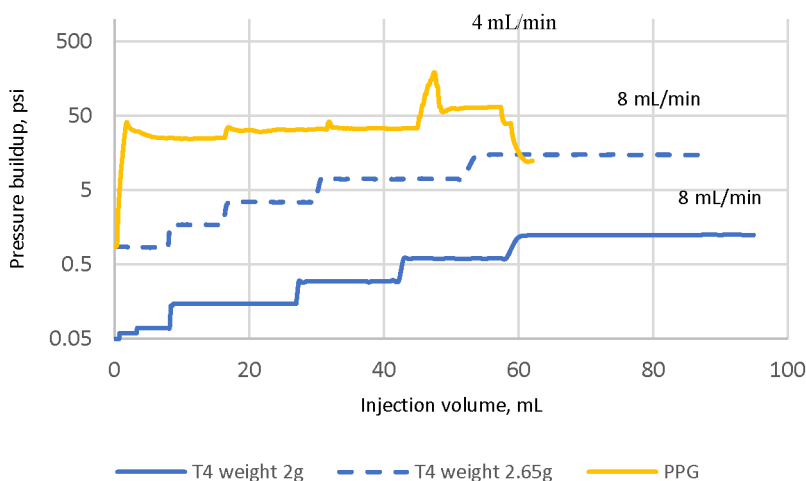
Figure 4—Pressure buildup during water flow through CO₂ plastic packed fracture

Figure 5 compares the stable pressure buildup during water flow through different CO₂ based plastic packed fracture. With same weight 2.65g and particle size range 40-80 mesh, sample T2 and T9 showed much higher-pressure buildup than other samples in 2mm fracture. Similar results were obtained at a loose pack weight 2.0g. The absolute densities of six samples were all 1.2 g/cm³. The difference came from the shape of the grinded particles. Microscope results on Figure 6 showed that grinded T2 and T9 had an irregular shape while other grinded samples had a smooth shape. These resulted in a larger apparent volume of T2 and T9 than other samples. Moreover, T2 and T9 were polyurethane with slight elastic deformation

abilities. With large apparent volume and deformation ability, the pack in the fracture was much firmer than other four samples.

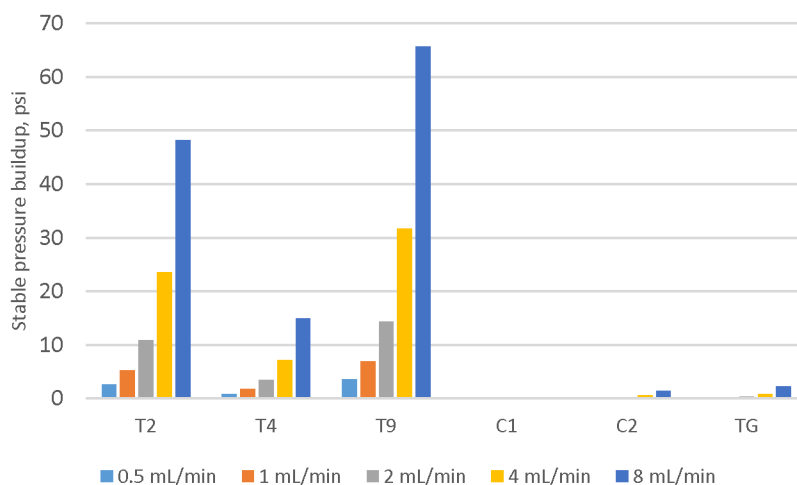


Figure 5—Stable pressure buildup during water flow through CO₂ plastic packed fracture with the same weight 2.65g

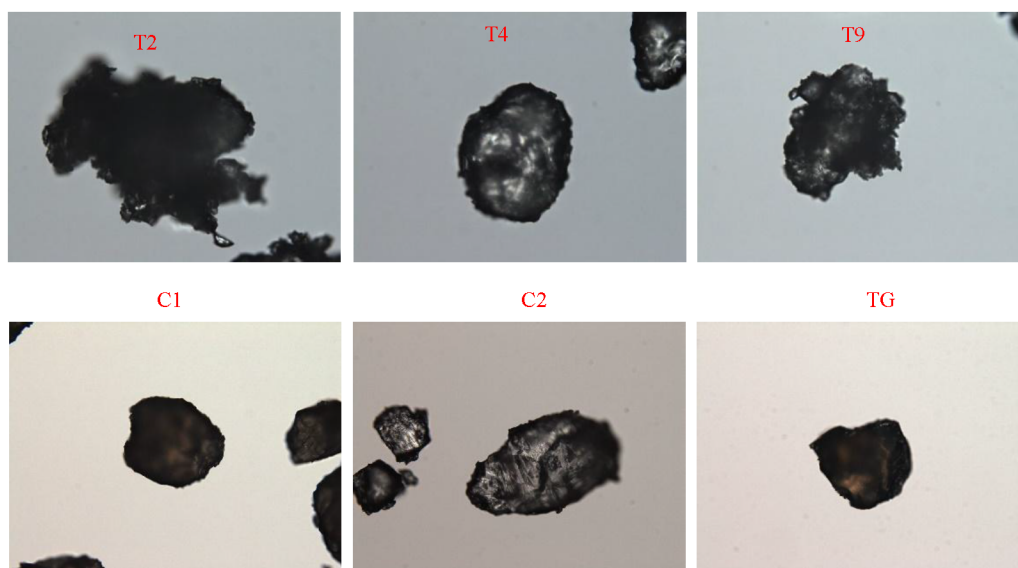


Figure 6—Shape of three grinded samples under microscope

Although T4, C1, C2 and TG had very similar apparent density, T4 showed much higher-pressure buildup than other 3 samples. The possible explanation was the melting problem. C1, C2 and TG were all pure PPC with much lower glass transition temperature than polyurethane T4. Figure 7 is the CO₂ plastic pack after plug tests. Sample C1 demonstrated obvious melting after test. This resulted in collapse of the pack and not fully fill the fracture. C2 and TG had similar problem but not as severe as C1. In comparison, T4 pack was not melted and easy grinded into the initial particle shape.

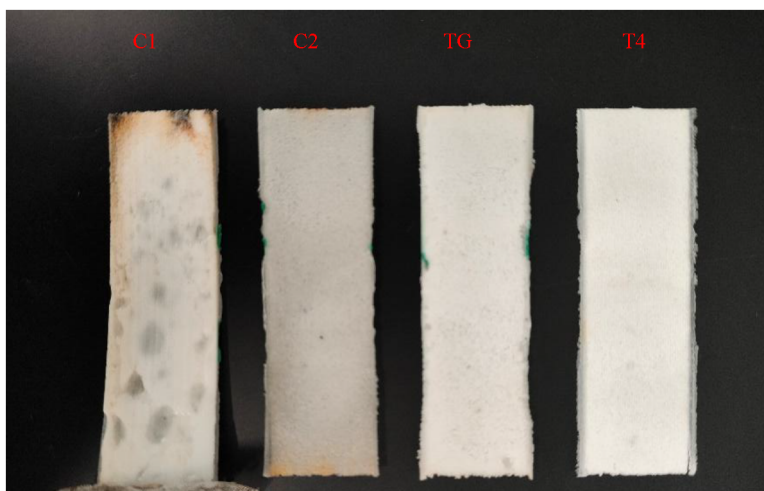


Figure 7—CO₂ plastic pack after plug tests

From the above test, temperature plays a critical role for the plugging performance of plastic. The effect of temperature on the plugging ability of T2 was tested. Besides temperature 95°C, tests at 110°C and 120°C were conducted. Figure 8 shows the stable pressures. With the increase of temperature, the pressure buildup was decreased. But the pack was still effective to the water flow.

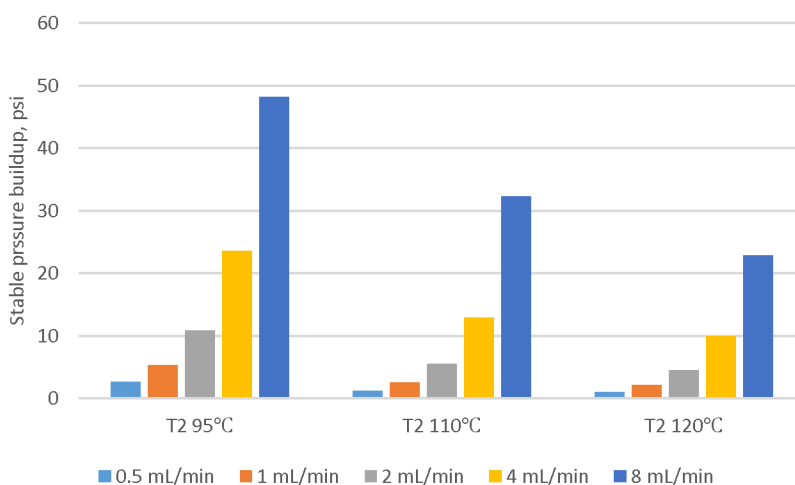


Figure 8—Stable pressure buildup during water flow through T2 packed fracture at different temperatures

Packing test

The packing tests simulated the injection process of CO₂ based plastic samples. Usually the injected particles were easily to be trapped at the position with abrupt flow rate change such as a contraction. With more particle trapped, a particle pack was formed. The flow was blocked accordingly with the injection pressure increase. The concentrations of CO₂ plastic were all 8000 mg/L suspended in 1000 mg/L polymer solution. The mesh sizes were all 40-80 mesh. Figure 9 shows the pressure change with the CO₂ plastic injection into a contraction fracture. Take sample T4 as example, the pressure did not increase at first 20 minutes injection. There was very little CO₂ plastic produced from the outlet, indicating that most of the injected particles was trapped at the contraction. But the fracture was not totally sealed with no effective blocking. After 20 minutes, the pressure buildup was increased slowly with time, indicating that the particles had packed across the whole fracture intersection. The pressure buildup increase was not smoothly but with abruptions. This was explained by a pack-break-pack process. The pack became unstable and broken when pressure above

some threshold pressure. With more particle trapped, the pack was stabilized again and withstood higher pressure. Sample T2 and T9 showed much quick pressure increasing and packing than other four samples. The possible reason was that the irregular shape of T2 and T9 make them easier to be carried and trapped during the injection. Figure 10 shows the packing of T2 in the contraction after test.

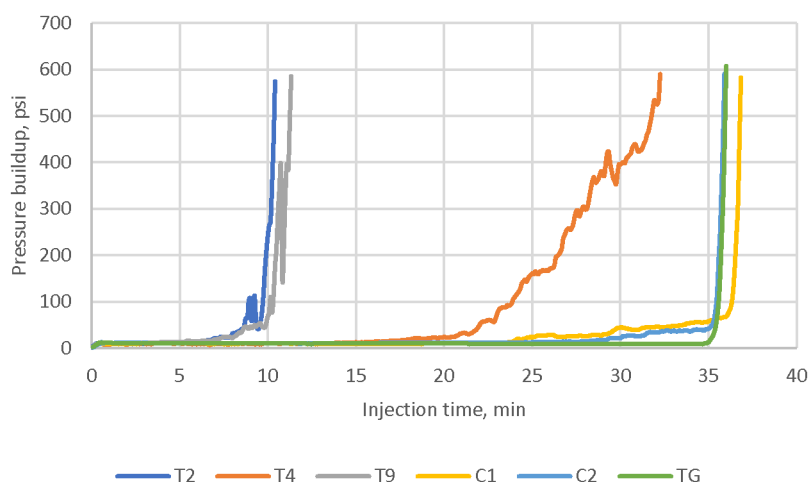


Figure 9—Pressure buildup during CO₂ plastic flow through contraction fracture



Figure 10—Packing of T2 in the contraction after test

After the injection pressure arrived at 4 Mpa (580 psi), the flow was switched to water injection to check the blocking ability of the in-situ formed pack to water flow. Figure 11 shows the stable pressures. Sample T2, T4 and T9 showed very similar pressure. This maybe because the tightness of the pack was similar at a fixed packing pressure. C1, C2 and TG failed to produce effective pressure for the water flow. Possible reason was that no effective packing was formed even after 35 minutes injection.

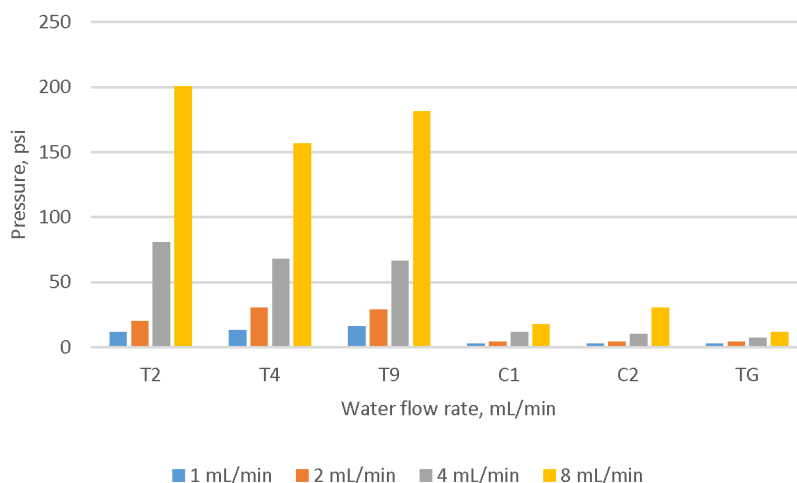


Figure 11—Stable pressure during water flow through in-situ formed pack

Because of the thermal plastic nature, the performance of CO₂ plastic at high temperature could deviate from the case at room temperature. Using sample T2 as candidate, the flow behavior was test at 95 °C with same procedures as above. Figure 12 shows the pressure change with the T2 injection into a contraction fracture at 95 °C. Two tests with different particle sizes were compared. In the test using 40-80 mesh T2, the injection pressure arrived at 4Mpa (580 psi) at 16.9 mins. This is much longer than the case of 10.4 min at room temperature. At high temperature, the T2 particles became deformed under pressure, making them easily passing through the contraction and delay the packing. A possible solution is to increase particle size. In the second test using mixture of 30-40 mesh and 40-80 mesh T2, the pressure increased quickly. Figure 13 shows the stable pressures during water flow through in-situ formed pack at 95°C. Although T2 of 40-80 mesh successfully packed at the contraction, the in-situ pack only produced very limited blocking to the subsequent water flow. As comparison, mixture of 30-40 mesh and 40-80 mesh T2 was much effective to block water flow.

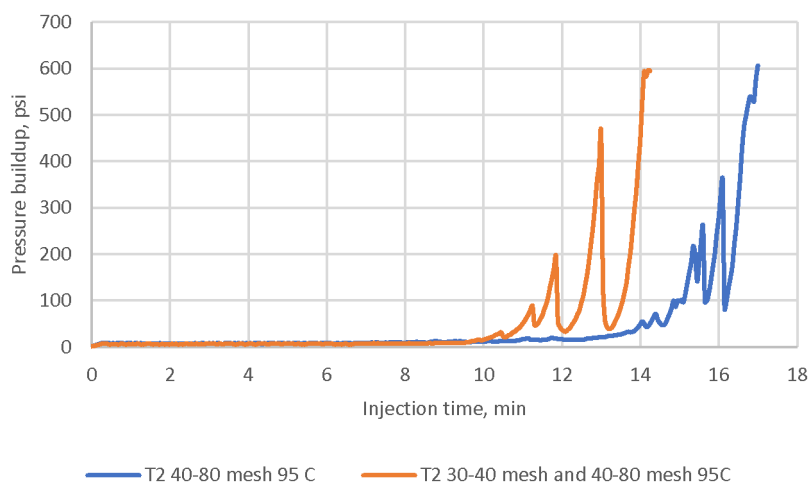


Figure 12—Pressure buildup during CO₂ plastic flow through contraction fracture 95#

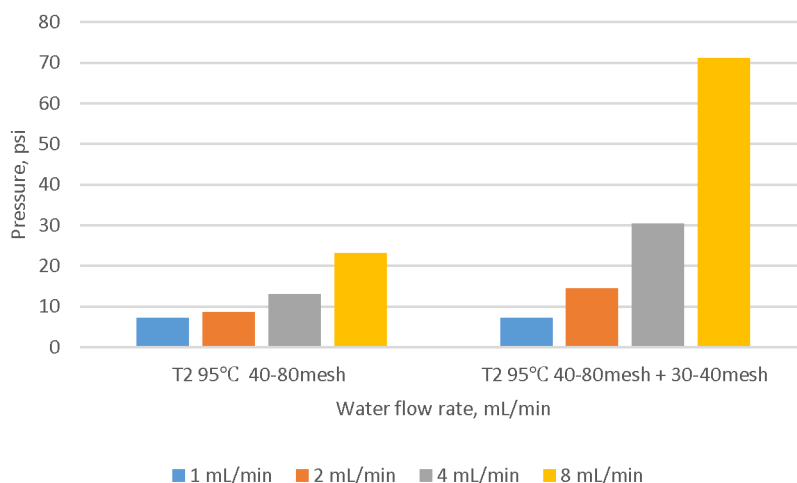


Figure 13—Stable pressure during water flow through in-situ formed pack at 95#

Conclusions

This work evaluated the blocking and packing behavior of crushed CO₂ based plastic samples in fractures. Totally 6 commercial samples available in the market including PPC based polyurethane T2, T4, T9 and PPC C1, C2, TG were compared. Below items summarize major findings:

1. The plastic samples need to be grinded by liquid nitrogen mill technology. All the grinded samples were oil-wet with contact angle to oil ranged from 56° to 75°. The apparent density of grinded T2 and T9 was much lower than other 4 samples.
2. Two samples T2, and T4 showed good thermal stability in brines at 95°C with almost no weight loss after aging for 6 months. Sample T9, C1 and C2 had weight loss around 10% to 30%.
3. In blocking tests, T2 and T9 pack in open fracture produced much higher-pressure buildup during water flow than that of other 4 samples at 95°C. The pressure decreased in tests at temperature 110°C and 120°C. C1, C2 and TG with low glass transition temperature partially melted at 95°C and showed very weak blocking.
4. In the packing tests, T2 and T9 injection showed much quicker packing and pressure buildup in a contraction fracture than other samples. The blocking ability of the in-situ formed pack to the following water flow was similar at a fixed packing pressure. C1, C2 and TG failed to produce effective pack in the contraction. At high temperature 95°C, the packing of T2 was delayed with a weak blocking to the subsequent flow. Increasing particle size was an effective way to solve this problem.

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